

Statistical Analysis and Reproducibility Appendix

Tail-sensitive HRR index in PMData

BoTao Cai

College of Physical Education, Jimei University, Xiamen 361021, China

Corresponding author: BoTao Cai; email: xcai2004xcai@gmail.com

S1. Purpose and analysis hierarchy

This appendix records the executable analysis specification and the reviewer-requested robustness analyses. The unit of independence was the participant. Sessions were repeated observations and were never randomly divided across training and test sets.

Supplementary Table S1. Analysis hierarchy

Level	Analysis	Interpretation
Principal in revised report	Participant-balanced held-out MAE: ten-bin tHRR-I versus linear decile score	Exploratory development-sample comparison; not preregistered
Secondary	Within-person Spearman association; comparison with intercept-only model	Ranking and absolute calibration
Exploratory	Alternative formula families, load correlations, high-RPE AUC, ordinal agreement, matching and anchor sensitivities	Hypothesis-generating only

The principal comparison was designated during revision. No timestamped analysis plan, confirmatory p values, or multiplicity-adjusted superiority claims were available.

S2. Session reconstruction and quality control

PMSys reports were converted to Europe/Oslo local time and linked one to one to Fitbit tracker sessions with compatible activity labels. Candidate reports could occur from 15 minutes before to 12 hours after the tracker session. Duration difference could not exceed max(10 minutes, 35% of reported duration). A greedy assignment minimized report delay and proportional duration difference. The primary set further required a bidirectionally unique match, an RPE delay of 15-180 minutes, valid RPE, at least 80% valid heart-rate coverage, and at least 10 valid minutes. Heart-rate observations required Fitbit confidence at least 2 and 30-220 beats·min⁻¹. Each observation was weighted by the following interval. Intervals were capped at 30 seconds so missing gaps did not contribute observed time. The absence of a shared PMSys-Fitbit session identifier means the reconstructed match is not a verified gold standard.

Supplementary Table S2. Session-matching stability

Rule	Selected pairs	Bidirectionally unique	Primary pairs retained
Primary 15-180-min window	449	443	255 (100.0%)
15-90-min window	401	398	225 (88.2%)
30-120-min window	413	408	232 (91.0%)

Rule	Selected pairs	Bidirectionally unique	Primary pairs retained
Primary window + duration difference ≤ 10 min	363	357	214 (83.9%)
Delay-priority matching cost	449	443	255 (100.0%)
Duration-priority matching cost	449	443	255 (100.0%)

Retention is relative to the 255 primary analysis pairs. Alternative windows change the eligible set; cost changes did not change primary pair identity.

S3. Physiological anchors and formulas

$$HRR(t) = \text{clip}([HR(t) - HR_{rest}] / [HR_{max} - HR_{rest}], 0, 1).$$

HR_{rest} was the participant median of Fitbit daily resting-heart-rate values, with a sleep-score value used when daily values were unavailable. Twelve participants had a PMData-provided HR_{max} of undocumented provenance. Three missing values were estimated as $208 - 0.7 \times \text{age}$. Exact session-date resting HR was evaluated separately when available.

$$tHRR-I(\lambda) = [\sum_i P_i c_i \exp(\lambda c_i)] / [\sum_i P_i \exp(\lambda c_i)], \lambda \geq 0.$$

$$\text{Continuous } tHRR-I(\lambda) = [\sum_t \Delta t h_t \exp(\lambda h_t)] / [\sum_t \Delta t \exp(\lambda h_t)].$$

$$\Delta \text{tilt}(\lambda) = tHRR-I(\lambda) - \mu HRR.$$

P_i is the time proportion in HRR bin i , c_i is its midpoint, and h_t is clipped continuous HRR. The ten-bin primary representation used $c_i = (i - 0.5) / 10$. The transform is bounded and monotonic in λ . At the full-development λ of 6.2, adjacent deciles differ 1.86-fold in relative weight and $HRR = 0.95$ receives 265-fold the pre-normalization weight of $HRR = 0.05$. Δtilt responds to dispersion and higher moments, including asymmetry and upper-tail exposure. It is not specific to discrete high-intensity events.

Comparator definitions were: linear decile score $= \sum_i P_i i$; mean $HRR = \sum_t \Delta t h_t / \sum_t \Delta t$; entropic $HRR = \log[\sum_i P_i \exp(\lambda c_i)] / \lambda$; power-mean $HRR = [\sum_i P_i c_i^q]^{1/q}$; original exponential score $= \sum_i P_i \exp(k i)$. Banister TRIMP was integrated from minute-level HRR with sex-specific coefficients and included duration.

S4. Nested evaluation and uncertainty

The outer loop held out one entire participant. Within each training set, λ was searched from 0.0 to 15.0 in steps of 0.1, q from 1.0 to 12.0 in steps of 0.1, and original k from 0.01 to 1.50 in steps of 0.01. The objective was the median within-person Spearman correlation among training participants with at least five sessions. The smallest parameter resolved ties. A participant-balanced linear calibration model then predicted the held-out participant, with each training participant contributing equal total weight.

Reported MAE and RMSE are participant-balanced. Pooled R^2 is included as a complementary calibration summary. The 95% intervals use 5000 participant-cluster bootstrap resamples with seed 20260729. They condition on the realized held-out predictions and do not rerun formula-family selection. A separate 200-split analysis with seed 20260731 randomly assigned 12 participants to training and three to test, and repeated λ selection and calibration for the locked tHRR-I family.

Supplementary Table S3. Discretization and resting-HR sensitivity

Estimator	Sessions/participants	Full λ	Outer λ median [IQR]	Balanced MAE	Median within-person rho
5 bins	255/15	12.6	12.6 [12.6 to 13.3]	1.284	0.708
10 bins (development candidate)	255/15	6.2	8.9 [6.2 to 9.3]	1.206	0.666
20 bins	255/15	6.3	7.4 [6.3 to 9.9]	1.223	0.644
Continuous time	255/15	6.1	6.1 [6.1 to 9.1]	1.239	0.644
10 bins + session-date HRrest	209/12	13.0	13.0 [9.2 to 13.0]	1.277	0.548

Parameters are not directly comparable across bin widths because the support points differ. The session-date HRrest analysis used the available subset.

Supplementary Table S4. Complete nested model performance after alternative session matching

Rule	Sessions/participants	Full λ	tHRR-I MAE	Linear MAE	MAE difference [95% conditional CI]
15-90-min report window	232/15	6.0	1.221	1.309	-0.089 [-0.149 to -0.029]
30-120-min report window	240/15	6.0	1.207	1.290	-0.083 [-0.151 to -0.024]
Duration difference ≤ 10 min	220/15	7.7	1.212	1.280	-0.068 [-0.130 to -0.009]
Delay-priority cost	264/15	9.3	1.202	1.297	-0.095 [-0.165 to -0.029]
Duration-priority cost	264/15	9.3	1.202	1.297	-0.095 [-0.163 to -0.026]

Each rule rebuilt the eligible session set and repeated participant-level parameter selection and calibration. Intervals remain conditional and exploratory.

Across 200 repeated 12/3 participant splits, median selected λ was 9.3 (IQR 8.9 to 9.3). Median tHRR-I minus linear MAE was -0.094 RPE units (IQR -0.135 to -0.038), and 86.0% of splits favored tHRR-I. These are partition-sensitivity summaries, not external-validation confidence intervals.

S5. Raw-signal and physiological-anchor sensitivities

Supplementary Table S5. Raw-signal and anchor sensitivity of nested prediction

Variant	Sessions/participants	Full λ	tHRR-I MAE	Linear MAE	MAE difference [95% conditional CI]
Rebuilt primary analysis	255/15	6.2	1.206	1.302	-0.097 [-0.163 to -0.029]
Winsorize upper 0.5% of session HR	255/15	6.2	1.203	1.302	-0.099 [-0.169 to -0.031]
Winsorize upper 1% of session HR	255/15	6.2	1.208	1.302	-0.094 [-0.158 to -0.028]
Cap HRR at 0.95	255/15	6.2	1.206	1.302	-0.097 [-0.161 to -0.030]
Cap sampling gaps at 15 s	255/15	6.2	1.206	1.302	-0.097 [-0.165 to -0.033]
Require Fitbit confidence=3	0/0	Not estimable	Not estimable	Not estimable	Not estimable
HRmax +5 beats·min ⁻¹	255/15	6.9	1.226	1.310	-0.083 [-0.146 to -0.019]
HRmax -5 beats·min ⁻¹	255/15	6.8	1.208	1.306	-0.098 [-0.155 to -0.040]
HRrest +3 beats·min ⁻¹	255/15	5.8	1.206	1.301	-0.096 [-0.162 to -0.029]
HRrest -3 beats·min ⁻¹	255/15	6.2	1.215	1.301	-0.086 [-0.152 to -0.023]
Age-predicted HRmax for all	255/15	4.6	1.374	1.406	-0.032 [-0.084 to 0.026]

The rebuilt baseline reproduced archived bin proportions to machine precision. No session met the primary 80% coverage rule when Fitbit confidence=3 alone was required. The all-Tanaka result shows that HRmax specification can materially affect calibration.

S6. Outcome-scale and high-RPE sensitivities

Supplementary Table S6. Exploratory outcome sensitivities

Analysis	Eligible data	tHRR-I	Linear	Difference
Participant-balanced high-RPE AUC	8 participants; 221 sessions; 50 high	0.817	0.673	0.144 [0.028 to 0.268]
Participant-specific quadratic-weighted kappa	9 participants	0.416	0.074	0.181

High RPE was defined post hoc as RPE ≥ 8 . AUC was calculated only for participants who had both classes. Rounded 1-10 predictions were used for quadratic-weighted kappa.

Session-RPE load equals RPE multiplied by duration. The objective load scores also include duration. Their association is therefore partly mathematically coupled and does not provide independent physiological validation.

S7. Clipping, prediction range, and safeguards

Before clipping, 0.034% of weighted valid time was below HRR=0 and 0.649% was above HRR=1. Any below-zero values occurred in 12 sessions and any above-one values in 22 sessions. Exact session-date resting HR was available for 209 sessions.

Supplementary Table S7. Held-out prediction range

Model	Minimum	Maximum	Outside archived 1-10 range
tHRR-I	2.850	7.169	0
Linear decile	3.389	7.100	0
Mean HRR	3.516	7.272	0
Intercept only	5.819	6.141	0

S8. Reproducibility specification

Required Python versions were Python 3.12.13, pandas 2.2.3, NumPy 2.3.5, openpyxl 3.1.5, and python-docx 1.2.0. Fixed random seeds are declared in each analysis script; reviewer_round3_analysis.py used seed 20260730.

Run order: (1) set PMDATA_ROOT to the licensed PMData directory; (2) run build_pmdata_session_dataset.py; (3) run develop_improved_formula.py; (4) run analyze_improved_formula_sensitivity.py; (5) run reviewer_revision_analysis.py; (6) run reviewer_round2_analysis.py; (7) run reviewer_round3_analysis.py; (8) run build_improved_figures.mjs when figure software is available; and (9) run the workbook, manuscript, supplement, and submission-document builders. The package does not redistribute raw PMData.

All derived CSV and JSON outputs used in the manuscript are archived under 04_Supplement/Derived_results. Source URLs, licensing information, and integrity manifests are archived under 05_Reproducibility. Paths in analysis scripts are relative or configurable; no author-specific path is required.

S9. Claims not resolved by internal reanalysis

The available public data cannot supply an independent external cohort, measured maximal heart rate for all participants, ECG-reference heart rate, a blinded hand-labeled PMSys-Fitbit match set, or a physiological outcome independent of RPE. The histogram-based index also cannot represent temporal ordering or recovery kinetics. These items must not be inferred from the present analyses. The manuscript therefore labels the study as exploratory method development and requires independent validation of the locked ten-bin formula with $\lambda=6.2$.